

A PROOF OF THE TWO-DIMENSIONAL GAUSSIAN-MOMENTS AND NULLCONE CONJECTURES, FORMALLY VERIFIED

ELIOTT CASSIDY

ABSTRACT. Let E be the Wick (Gaussian) expectation on $\mathbb{C}[Z, W]$ determined by $E[Z^a W^b] = a! [a = b]$, the moment functional of a single complex Gaussian. The *Gaussian-Moments Conjecture* of Derksen, van den Essen and Zhao asserts that the kernel of E is a Mathieu-Zhao space; in n variables it implies the Jacobian Conjecture, and it is false for $n \geq 3$. We prove its sharp surviving case $n = 2$: if $E(P^m) = 0$ for all $m \geq 1$ then $E(Q P^m) = 0$ for all large m . The proof passes through the two-dimensional *Nullcone Conjecture*, that a moment-null polynomial has one-sided charge, and reduces the residual analytic content to the one-variable Duistermaat-van der Kallen constant-term theorem. We give a self-contained, valuation-free proof of the latter through a Galois orbit-product on the splitting field of $X^M - tR$, fed by the Weierstrass small-root product $\Pi = ct$, whose analytic heart $\partial_t h(0, t) = 0$ is proved inside a single power-series frame. The entire argument has been formalized in the Lean 4 proof assistant on top of `mathlib`; the top-level theorem is machine-checked and depends only on the three foundational axioms of that library.

1. INTRODUCTION

1.1. The conjecture. Fix the polynomial ring $\mathbb{C}[Z, W]$ in two variables and regard $W = \bar{Z}$ as the conjugate of a single complex Gaussian variable Z . The associated *Wick*, or *Gaussian*, expectation is the linear functional

$$E: \mathbb{C}[Z, W] \longrightarrow \mathbb{C}, \quad E[Z^a W^b] = a! [a = b], \quad (1)$$

where $[a = b]$ is 1 if $a = b$ and 0 otherwise. Equivalently, writing $Z = x + iy$, $W = x - iy$, E is integration against the standard planar Gaussian, and (1) is the classical Wick formula for its moments. The functional E annihilates every monomial off the diagonal, i.e. every monomial of nonzero *charge*

$$\text{ch}(Z^a W^b) = a - b.$$

A linear subspace $\mathcal{M}$ of a commutative $\mathbb{C}$ -algebra A is a *Mathieu-Zhao space* (Zhao [4]; originally *Mathieu subspace*) if, whenever $a \in A$ satisfies $a^m \in \mathcal{M}$ for all $m \geq 1$, one has $ba^m \in \mathcal{M}$ for all sufficiently large m and every $b \in A$. These spaces isolate the algebraic content of a family of conjectures descending from the Jacobian Conjecture: Mathieu's conjecture on integrals of G -finite functions [3], Zhao's Image Conjecture [5, 6], and their common two-variable Gaussian shadow.

Conjecture 1.1 (Gaussian-Moments Conjecture, [1]). *For every n , the kernel $\ker E_n$ of the n -dimensional Wick expectation is a Mathieu-Zhao space of $\mathbb{C}[Z_1, W_1, \dots, Z_n, W_n]$.*

Derksen, van den Essen and Zhao [1] introduced Conjecture 1.1, proved that it implies the Jacobian Conjecture, and exhibited a counterexample to a broader moment-vanishing statement showing that the conjecture is *false* for $n \geq 3$. Dimension two is thus the sharp surviving case. Our first result is that it survives.

Theorem 1.2 (GMC(2)). *Let $P, Q \in \mathbb{C}[Z, W]$. If $E(P^m) = 0$ for all $m \geq 1$, then there is N such that $E(Q P^m) = 0$ for all $m \geq N$.*

Unwinding (1), the obstruction is purely one of *charge geometry*. For a polynomial P let $\text{supp } P \subset \mathbb{N}^2$ be its exponent support and $\text{ch}(s) = s_0 - s_1$ the charge of a monomial.

Theorem 1.3 (NC(2), the two-dimensional Nullcone Conjecture). *Let $P \in \mathbb{C}[Z, W]$. If $E(P^m) = 0$ for all $m \geq 1$, then P has one-sided charge: either $\text{ch}(s) \geq 1$ for every $s \in \text{supp } P$, or $\text{ch}(s) \leq -1$ for every $s \in \text{supp } P$.*

Theorem 1.3 is the structural heart of Theorem 1.2: it says that the only way for all central powers of P to be moment-null is for P to be strictly one-sided (no monomial can have charge 0, and all charges must agree in sign). The passage $\text{NC}(2) \Rightarrow \text{GMC}(2)$ is elementary (Section 2). The content is Theorem 1.3 itself.

1.2. Method. The proof of Theorem 1.3 confronts one, and only one, genuinely analytic input: the one-variable *Duistermaat–van der Kallen theorem* on constant terms in powers of a Laurent polynomial [2]. In the exact form we need it reads:

Theorem 1.4 (Duistermaat–van der Kallen, finite-support form). *Let $f(z) = \sum_i c_i z^{q_i}$ be a Laurent polynomial with distinct exponents $q_i \in \mathbb{Z}$ and nonzero coefficients $c_i \in \mathbb{C}$, whose support straddles zero: either 0 occurs among the q_i , or both a negative and a positive exponent occur. Then there is $m \geq 1$ with $\text{CT}(f^m) \neq 0$, where CT denotes the constant Laurent coefficient.*

The classical theorem of Duistermaat and van der Kallen [2] classifies the Laurent polynomials all of whose positive powers have vanishing constant term; Theorem 1.4 is the contrapositive in the two-sided case, together with the trivial zero-charge case. We do not import the analytic proof of [2]. Instead we give a self-contained algebraic proof (Sections 4–6) that runs as follows. Clearing denominators turns f into a genuine polynomial and reduces the vanishing of all $\text{CT}(f^m)$ to the vanishing of the central coefficients $D_m = [X^{Mm}]R(X)^m$ of a single polynomial R ; these are precisely the coefficients governing the small roots of

$$\Phi = X^M - tR(X) \in \mathbb{C}(t)[X].$$

Over the splitting field of Φ , a transitive Galois action and a Vieta identity force a contradiction *as soon as* the product of the M small roots equals ct with $c \neq 0$ (Section 5). That product identity is exactly the Weierstrass small-root product $\Pi = ct$, whose only nontrivial ingredient is the scalar analytic identity $\partial_t h(0, t) = 0$ under $D_m \equiv 0$, proved inside the single power-series frame $(\mathbb{C}((x)))[[t]]$ (Section 6). No extension of a valuation to an algebraic closure, and no Puiseux expansion, is required.

1.3. Formalization. Every step above — Theorems 1.2, 1.3, 1.4, the univariate reduction, the orbit-product contradiction, and the Weierstrass analysis — has been formalized in the Lean 4 proof assistant on top of `mathlib` [8]. The top-level statement is

```
theorem GMC2.gmc2 (P Q : MvPolynomial (Fin 2) ℂ)
  (hnull : ∀ m, 1 ≤ m → E (P ^ m) = 0) :
  ∃ N, ∀ m ≥ N, E (Q * P ^ m) = 0
```

and a kernel audit reports

```
#print axioms GMC2.gmc2 = [propext, Classical.choice, Quot.sound],
```

the three standard axioms of `mathlib`: there is no `sorry`, no appeal to `native_decide`, and no additional axiom anywhere in the transitive dependency. In particular the one-variable input, Theorem 1.4, is itself formalized rather than assumed. The proof is described module by module in Section 7.

1.4. Organization. Section 2 records the functional E and the elementary reduction $\text{NC}(2) \Rightarrow \text{GMC}(2)$. Section 3 reduces $\text{NC}(2)$ to Theorem 1.4. Section 4 reduces Theorem 1.4 to a single polynomial statement (the *small-root packet product*). Section 5 closes that statement by a Galois orbit-product. Section 6 supplies the Weierstrass small-root product and its analytic core. Section 7 documents the Lean formalization and its verification. Section 8 collects remarks on sharpness and on the Jacobian Conjecture.

2. THE WICK FUNCTIONAL AND THE CHARGE REDUCTION

Write monomials of $\mathbb{C}[Z, W]$ as $Z^{s_0}W^{s_1}$ with $s = (s_0, s_1) \in \mathbb{N}^2$, and set

$$\text{wt}(s) = \begin{cases} s_0! & s_0 = s_1, \\ 0 & \text{otherwise,} \end{cases} \quad E(P) = \sum_{s \in \text{supp } P} P_s \text{wt}(s),$$

where P_s is the coefficient of $Z^{s_0}W^{s_1}$ in P . Thus E agrees with (1) and annihilates every monomial of nonzero charge $\text{ch}(s) = s_0 - s_1$.

Lemma 2.1. *If $P, R \in \mathbb{C}[Z, W]$ have the property that every monomial of P and every monomial of R has charge ≥ 1 (respectively ≤ -1), then $E(R) = 0$; more generally $E(PR) = 0$ whenever every monomial of PR has nonzero charge.*

Proof. Charge is additive under multiplication, $\text{ch}(s+t) = \text{ch}(s) + \text{ch}(t)$, so every monomial of a product of strictly one-sided polynomials has charge of the same strict sign, hence nonzero; and wt vanishes off charge 0. $\square$

Proposition 2.2 ($\text{NC}(2) \Rightarrow \text{GMC}(2)$). *Assume Theorem 1.3. Then Theorem 1.2 holds; explicitly, if $E(P^m) = 0$ for all $m \geq 1$ then $E(Q P^m) = 0$ for all $m \geq \deg_W Q + 1$ (in the case of positive charge; symmetrically for negative charge).*

Proof. By Theorem 1.3, P is strictly one-sided; say every monomial of P has charge ≥ 1 (the other case is symmetric under $Z \leftrightarrow W$). Every monomial of P^m then has charge $\geq m$. A monomial $Z^a W^b$ of Q has $b \leq \deg_W Q$, so its charge is $\geq -\deg_W Q$; hence a monomial of $Q P^m$ has charge $\geq m - \deg_W Q \geq 1$ once $m \geq \deg_W Q + 1$. By Lemma 2.1, $E(Q P^m) = 0$ for all such m . $\square$

The threshold $N = \deg_W Q + 1$ is explicit and uniform: the whole difficulty is concentrated in Theorem 1.3, to which we now turn.

3. REDUCTION OF $\text{NC}(2)$ TO THE CONSTANT-TERM THEOREM

Suppose $E(P^m) = 0$ for all $m \geq 1$ but, for contradiction, that P is *not* one-sided: its charge support is not contained in $\mathbb{Z}_{\geq 1}$ nor in $\mathbb{Z}_{\leq -1}$. Because E only sees the diagonal, only the *charge-balanced* part of each power matters:

$$E(P^m) = \sum_{\substack{r \in \mathbb{N}^{\text{supp } P} \\ |r|=m, \sum_i r_i \text{ch}(s_i)=0}} \binom{m}{r} \left(\prod_i P_{s_i}^{r_i} \right) (\text{diagonal Wick weight}), \quad (2)$$

the sum over compositions r of m across the monomials s_i of P whose total charge is zero. Passing to the associated one-dimensional charge data $q_i = \text{ch}(s_i) \in \mathbb{Z}$ with coefficients $c_i = P_{s_i} \in \mathbb{C}^\times$ isolates the *angular* content of (2): the balanced compositions of $q = (q_i)$ are exactly the constant-term channels of the Laurent polynomial $f(z) = \sum_i c_i z^{q_i}$. The radial content — the factorials wt and the Frobenius amplification that upgrades a single balanced face to all of them

— is carried by an elementary but non-vacuous combinatorial argument (the lowest-balanced-face / Kummer-carry mechanism, formalized in the module `GMC2NC2`) whose only external call is the following: the constant-term relations of f cannot all vanish.

Concretely, let $q: \{1, \dots, k\} \rightarrow \mathbb{Z}$ be injective with values the distinct charges and $c_i \neq 0$ the corresponding coefficients. For $m \geq 0$ define the universal constant-term polynomial

$$\mathcal{C}_m(q) = \sum_{\substack{r \in \mathbb{N}^k, |r|=m \\ \sum_i r_i q_i = 0}} \binom{m}{r} X^r \in \mathbb{Q}[X_1, \dots, X_k], \quad (3)$$

so that $\mathcal{C}_m(q)(c) = \text{CT}(f^m)$. The hypothesis $E(P^m) = 0$ for all m forces, after the Frobenius reduction, $\mathcal{C}_m(q)(c) = 0$ for all $m \geq 1$. Since P is two-sided, the charge set $\{q_i\}$ straddles zero. Theorem 1.4 then provides $m \geq 1$ with $\mathcal{C}_m(q)(c) \neq 0$, a contradiction. Hence P is one-sided, proving Theorem 1.3. $\square$

Remark 3.1. The zero-charge case of Theorem 1.4 is immediate: if some $q_{i_0} = 0$, then at $m = 1$ the single composition supported at i_0 is the unique balanced channel and $\mathcal{C}_1(q)(c) = c_{i_0} \neq 0$. The content is therefore the two-sided, no-zero case, treated next.

4. THE UNIVARIATE REDUCTION: FROM CONSTANT TERMS TO A SMALL-ROOT PACKET

Fix injective q straddling zero with no zero charge, and coefficients $c_i \neq 0$. Let $M = -\min_i q_i > 0$ and form the genuine polynomial

$$R(X) = \sum_i c_i X^{q_i + M} \in \mathbb{C}[X].$$

The shift is chosen so that $R(0) = c_{i_0} \neq 0$ (the unique minimal charge) and $M < \deg R$ (the unique maximal charge is strictly positive). A direct expansion identifies the constant-term relations (3) with central coefficients of powers of R :

$$[X^{Mm}] R(X)^m = \mathcal{C}_m(q)(c) \quad (m \geq 0). \quad (4)$$

Thus assuming $\mathcal{C}_m(q)(c) = 0$ for all $m \geq 1$ is the same as assuming $D_m := [X^{Mm}] R^m = 0$ for all $m \geq 1$.

These central coefficients are the moments of the small roots of

$$\Phi = X^M - tR(X) \in \mathbb{C}(t)[X], \quad (5)$$

a polynomial of degree $N = \deg R > M$ in X over the rational function field $\mathbb{C}(t)$. Modulo t it degenerates to X^M , so exactly M of its N roots are “small” (they tend to 0 as $t \rightarrow 0$); their product carries a t -adic valuation equal to 1 when the moments D_m vanish. The whole of Theorem 1.4 reduces to the following single statement about Φ .

Proposition 4.1 (Small-root packet product). *Let $R \in \mathbb{C}[X]$ with $1 \leq M < \deg R$ and $R(0) \neq 0$, and suppose $D_m = [X^{Mm}] R^m = 0$ for all $m \geq 1$. Then, in the splitting field L of $\Phi = X^M - tR$ over $\mathbb{C}(t)$, there is a finite set S of roots of Φ whose product equals ct for some $c \in \mathbb{C}^\times$:*

$$\prod_{\beta \in S} \beta = ct \in \mathbb{C}(t) \subset L.$$

Granting Proposition 4.1, Theorem 1.4 follows: were $D_m = 0$ for all $m \geq 1$, the packet product would be ct , and Section 5 shows this is impossible; hence some $D_m \neq 0$, i.e. some $\mathcal{C}_m(q)(c) \neq 0$.

5. THE GALOIS ORBIT-PRODUCT CONTRADICTION

We first record why a packet product equal to ct is impossible, independently of *how* the packet arises. This is the algebraic core, an orbit-product identity for the Galois action on the roots.

Lemma 5.1. $\Phi = X^M - tR$ is irreducible over $\mathbb{C}(t)$ whenever $1 \leq M$ and $R(0) \neq 0$.

Sketch. Φ is linear in t with content 1; the coefficients X^M and $-R$ of t^0, t^1 are coprime in $\mathbb{C}[X]$ because $R(0) \neq 0$ (so $X \nmid R$). Hence Φ is irreducible in $\mathbb{C}[X][t] = \mathbb{C}[t][X]$, and by Gauss's lemma over the fraction field $\mathbb{C}(t)$. (In the formalization this is carried out through the Bivariate swap $\mathbb{C}[t][X] \cong \mathbb{C}[X][t]$ and `mathlib`'s primitive/Gauss machinery.) $\square$

Let L be the splitting field of Φ over $F := \mathbb{C}(t)$ and $G = \text{Gal}(L/F)$. Since Φ is irreducible, G acts transitively on the set Ω of its roots. The key point is that the orbit-product identity below holds for an *arbitrary* G -stable factor of the product — one need not know that S is the “valuation-positive” packet, only that its product is G -fixed.

Lemma 5.2 (Orbit-product). *Let G act transitively on a finite set Ω with an equivariant embedding $f: \Omega \rightarrow L$, and let $S \subseteq \Omega$ be any subset such that $\pi := \prod_{\beta \in S} f(\beta)$ is G -fixed. Then for a basepoint $x \in \Omega$,*

$$\pi^{|G|} = \left(\prod_{\alpha \in \Omega} f(\alpha) \right)^{|S| \cdot |\text{Stab}_G(x)|}.$$

Proof. Applying every $g \in G$ and multiplying, $\prod_{g \in G} g \cdot \pi = \pi^{|G|}$ since π is fixed. On the other hand $\prod_{g \in G} g \cdot \pi = \prod_{\beta \in S} \prod_{g \in G} f(g\beta)$, and by transitivity $\prod_{g \in G} f(g\beta) = \left(\prod_{\alpha \in \Omega} f(\alpha) \right)^{|\text{Stab}_G(x)|}$ for every β . Taking the product over the $|S|$ elements of S gives the claim. $\square$

Proposition 5.3 (No packet has product ct). *With $F = \mathbb{C}(t)$, L , G as above, suppose $S \subseteq \Omega$ has G -fixed product $\prod_{\beta \in S} \beta = ct$ with $c \in \mathbb{C}^\times$. Then a contradiction ensues.*

Proof. Because $ct \in F$ lies in the base field, it is automatically G -fixed, so Lemma 5.2 applies with $\pi = ct$. The full product of the roots is, by Vieta on (5),

$$\prod_{\alpha \in \Omega} \alpha = (-1)^N \frac{[X^0]\Phi}{\text{lc}_X \Phi} = (-1)^N \frac{-tR(0)}{-t \text{lc } R} = (-1)^N \frac{R(0)}{\text{lc } R} =: d \in \mathbb{C}^\times,$$

a nonzero *constant*: the factor t cancels between the constant and leading coefficients of Φ . Substituting into Lemma 5.2,

$$(ct)^{|G|} = d^{|S| \cdot |\text{Stab}_G(x)|} \quad \text{in } F = \mathbb{C}(t).$$

The left side has t -adic valuation $|G| \geq 1$; the right side is a nonzero constant of valuation 0. A degree- $|G|$ monomial in t cannot equal a constant, so this is absurd. $\square$

Proposition 5.3 is what makes the reduction *valuation-free*. Nothing about the size of the roots is used; the only inputs are transitivity (Lemma 5.1), Vieta, and the elementary fact that a monomial is not a constant. It remains to produce a packet with product ct , i.e. to prove Proposition 4.1. This is where the Weierstrass analysis enters, and it is the only place any analysis is used.

6. THE WEIERSTRASS SMALL-ROOT PRODUCT AND ITS ANALYTIC CORE

Regard Φ as a polynomial in X over the complete local ring $A = \mathbb{C}[[t]]$:

$$\Phi = X^M - tR(X) \in A[X] \subset A[[X]].$$

Its reduction mod t is X^M , so Weierstrass preparation over $A[[X]]$ yields a factorization

$$\Phi = P \cdot h, \quad P \in A[X] \text{ monic of degree } M, \quad P \equiv X^M \pmod{t}, \quad h \in A[[X]]^\times. \quad (6)$$

P is the *distinguished* (small-root) factor; its M roots are the small roots of Φ , and the small-root product is $\Pi = (-1)^M P.\text{coeff}_0$.

6.1. From power series to a polynomial factorization. The factorization (6) lives in the power-series ring; to feed the packet into Section 5 we need P to divide Φ as *polynomials*, so that the roots of P are genuinely roots of Φ in L .

Lemma 6.1. *P divides Φ in the polynomial ring $A[X]$.*

Proof. Since P is monic, ordinary polynomial division gives $\Phi = P \cdot Q + r$ with $\deg_X r < M$. Coercing to $A[[X]]$ and comparing with (6), both $(h, 0)$ and (Q, r) are Weierstrass divisions of Φ by P ; by uniqueness of Weierstrass division over the Hausdorff ring A (the coefficient $P.\text{coeff}_0 = ct$ lies in the maximal ideal, and $P \equiv X^M \pmod{t}$ makes P a valid Weierstrass divisor), the remainder r vanishes. Hence $P \mid \Phi$ in $A[X]$. $\square$

Mapping $A = \mathbb{C}[[t]] \hookrightarrow \mathbb{C}((t))$ and then into $\Omega = \overline{\mathbb{C}((t))}$, and using that the polynomial (5) over $\mathbb{C}(t)$ coincides with its image over $\mathbb{C}((t))$, one obtains $P \mid \Phi$ over Ω . The splitting field L of $\Phi/\mathbb{C}(t)$ embeds $\mathbb{C}(t)$ -linearly into Ω (both split Φ), and the packet

$$S = \{\beta \in \Phi.\text{rootSet}(L) : \psi(\beta) \text{ is a root of } P\}, \quad \psi: L \hookrightarrow \Omega,$$

has, by Vieta on the monic P ,

$$\psi\left(\prod_{\beta \in S} \beta\right) = \prod (\text{roots of } P) = (-1)^M P.\text{coeff}_0.$$

Since ψ is injective and fixes $\mathbb{C}(t)$, the identity descends to L : $\prod_{\beta \in S} \beta = (-1)^M P.\text{coeff}_0$ once the latter is shown to be a base-field element. This is the content of the next, and final, computation.

6.2. The scalar identity $\Pi = ct$. Taking the X^0 -coefficient of (6) gives, with $h(0, t) := [X^0]h \in A^\times$ and $r_0 = R(0)$,

$$P.\text{coeff}_0 \cdot h(0, t) = [X^0]\Phi = -tr_0, \quad (7)$$

so $\Pi = (-1)^M P.\text{coeff}_0$ equals ct with $c = (-1)^{M+1} r_0$ if and only if $h(0, t) = 1$. The unit $h(0, t) \in \mathbb{C}[[t]]$ is normalized by $h(0, 0) = 1$ (from $P \equiv X^M \equiv \Phi \pmod{t}$ and cancellation of X^M in the domain $\mathbb{C}[[X]]$). It remains to show that its t -derivative vanishes:

$$\partial_t h(0, t) = 0 \quad \text{when } D_m = 0 \text{ for all } m \geq 1. \quad (8)$$

Over a field of characteristic 0, (8) together with $h(0, 0) = 1$ forces $h(0, t) = 1$, hence $\Pi = ct$, hence Proposition 4.1.

Identity (8) is the sole analytic ingredient of the entire paper, and it is a one-line logarithmic-derivative computation once the objects are placed in a *single* completion. Work in

$$\mathcal{A} = (\mathbb{C}((x)))[[t]],$$

the t -power series over the Laurent field $\mathbb{C}((x))$. Here $\Phi = x^M - tR(x)$, P and h all map in and are *units* (their t -constant terms are $x^M, x^M, 1$, all nonzero in the field $\mathbb{C}((x))$), so

R/Φ , $\partial_t P/P$, $\partial_t h/h$ are honest elements of $\mathcal{A}$. Differentiating $\Phi = P \cdot h$ in t and dividing by Φ ,

$$-\frac{R}{\Phi} = \frac{\partial_t P}{P} + \frac{\partial_t h}{h}.$$

Apply the $\mathbb{C}[[t]]$ -linear functional $[x^0]$ (the x^0 -Laurent coefficient, a ring homomorphism on the “disk” subring of series with nonnegative x -support). Three evaluations finish the argument:

- (i) $[x^0](\partial_t P/P) = 0$: P is monic of x -degree M and $\partial_t P$ has x -degree $< M$, so $\partial_t P/P$ has strictly negative x -degree and no x^0 term (a reversed-polynomial degree count);
- (ii) $[x^0](-R/\Phi) = -\sum_{m \geq 1} D_m t^{m-1}$: expanding $x^M/\Phi = (1 - tR/x^M)^{-1}$ as a geometric series identifies the x^0 coefficient with the moment generating function $\sum_m D_m t^m$;
- (iii) $[x^0](\partial_t h/h) = \partial_t h(0, t)/h(0, t)$: h and $\partial_t h$ have nonnegative x -support, so on that subring $[x^0]$ is the ring homomorphism $h \mapsto h(0, t)$, which commutes with ∂_t .

Combining, $-\sum_{m \geq 1} D_m t^{m-1} = 0 + \partial_t h(0, t)/h(0, t)$. Under $D_m \equiv 0$ the left side is 0, so $\partial_t h(0, t) = 0$, proving (8) and completing the proof of Proposition 4.1, hence of Theorems 1.4, 1.3, and 1.2. $\square$

Remark 6.2. The passage between the “space” completion $(\mathbb{C}((x)))[[t]]$ used for (8) and the “deformation” completion $\mathbb{C}[[t]][X]$ used for Lemma 6.1 is a transposition of the two power-series variables; both are algebraic and neither requires extending a valuation to the algebraic closure of $\mathbb{C}((t))$. This is the decisive simplification over the Puiseux/Newton-polygon route: the orbit-product contradiction (Proposition 5.3) accepts *any* packet whose product is ct , so the packet may be defined algebraically as the roots of the Weierstrass factor P rather than by valuation.

7. THE LEAN FORMALIZATION

The proof is formalized in Lean 4 against `mathlib` [8]. We summarize the module structure; the correspondence to the sections above is exact.

- **The functional and the charge reduction.** `E`, `wt`, `charge`, `NC2`, and the reduction `gmc2_of_nc2` (Proposition 2.2) are in `GMC2Reduction`. The threshold $N = \deg_W Q + 1$ is explicit.
- **`NC(2)  $\Leftarrow$  DvdK(1)`.** The lowest-balanced-face/Frobenius argument of Section 3 is `GMC2NC2` (with the normalized specialization contradiction `normalized_specialization_contradiction` and the discharged height witness), yielding `nc2_of_dvdK1` and `gmc2_of_dvdK1`.
- **The constant-term interface.** `DvdK1` (Theorem 1.4), `ChargesStraddleZero`, and the universal $\mathcal{C}_m(q)$ (3) (`constantTermRelation`) are in `GMC2DvdKInterface`, `GMC2LowestFaceExistence`, `GMC2ConstantTermRelations`; the zero-charge case and the both-signs reduction are in `GMC2DvdKZeroCharge`.
- **Univariate reduction.** `SinglePolyCrux` (Proposition 4.1), the shift $R = \text{shiftedPolynomial}$, the identity (4) (`GMC2LaurentShiftCheckA`), and the capstone `gmc2_of_crux` are in `GMC2DvdKUnivariateReduction`.
- **Orbit-product.** Lemma 5.2 and Proposition 5.3 are `GMC2OrbitProduct` and `GMC2Thm2067Wrapper` (`thm2067_contradiction`); irreducibility (Lemma 5.1) is `GMC2PhiIrreducible/GMC2DvdKAssembly`; Vieta ($\prod \alpha = d$) is `GMC2PhiVieta` (`prod_rootSet_Phi`); the char-0 specialization discharging separability and Galois-fixedness is `GMC2Thm2067HSonly` (`thm2067_reduced_to_hS`).
- **The frame bridge.** The valuation-free packet transport of Section 6 — root transport, the packet bijection, and the assembly $\prod_{\beta \in S} \beta = \text{algebraMap}(ct)$ — is `GMC2FrameBridgeRoots`, `GMC2FrameBridgePacket` (`exists_packet_prod_eq`, a `Finset.prod_bij`), `GMC2FrameBridge`, `GMC2FrameBridgeAssembly`.

- **Weierstrass and the analytic core.** (6), $P = \text{smallRootFactor}$, and (7) are `GMC2DvdKWeierstrass` (built on `mathlib`’s Weierstrass preparation); Lemma 6.1 is `GMC2FrameBridgeDvd` (`smallRootFactor_dvd.PhiPoly`); the normalization $h(0,0) = 1$ is `GMC2DvdKUnitOrigin`; the log-derivative identity (8) is proved in the unified frame $(\mathbb{C}((x)))[[t]]$ across `GMC2DvdKFrame`, `GMC2DvdKTranspose`, `GMC2DvdKHderivAssembly`, `GMC2DvdKTransposeAssembly`; the char-0 “vanishing derivative $\Rightarrow$ constant” step is `GMC2DvdKCharZeroClosing`.
- **Discharge of the crux and the front door.** The wiring over $\Omega = \overline{\mathbb{C}((x))}$ that proves `singlePolyCrux_holds`, and the unconditional theorem `gmc2_unconditional = gmc2_of_crux singlePolyCrux_holds`, are in `GMC2DvdKOmegaWiring`; the documented statement `GMC2.gmc2` is in `GMC2Main`.

Verification. The build target `GMC2Main` compiles the entire proof and is self-contained. The kernel axiom audit

```
#print axioms GMC2.gmc2 = [propext, Classical.choice, Quot.sound]
```

certifies that the theorem uses only `mathlib`’s three foundational axioms: there is no `sorry`, no `native_decide` (hence no unaudited kernel reduction), and no additional axiom. Every module terminates in a `#print axioms` directive with the same result, so the property holds throughout the dependency graph, not merely at the root.

8. CONCLUDING REMARKS

Sharpness. Theorem 1.2 is dimension-optimal: Derksen, van den Essen and Zhao [1] show the Gaussian-Moments Conjecture fails for $n \geq 3$. The obstruction to higher dimensions is genuinely multi-radial: the single hyperbolic charge $\text{ch} = s_0 - s_1$ of $\mathbb{C}[Z, W]$ is replaced by a lattice of charges, and one-sidedness is no longer forced. Theorem 1.3 is the exact statement that survives at $n = 2$: instability of the moment functional (all powers null) coincides with algebraic instability (a strict one-sided charge, a destabilizing one-parameter subgroup of the hyperbolic torus).

The one-variable input. The argument isolates the entire transcendental difficulty of the problem in the single scalar identity (8), a formal logarithmic derivative. This is a constructive counterpart to the analytic proof of Duistermaat and van der Kallen [2], replacing residues and monodromy of algebraic functions by a Weierstrass factorization and a Galois orbit-product, and in particular giving a proof of Theorem 1.4 that is elementary enough to be checked by a proof assistant.

Relation to the Jacobian Conjecture. By [1], the Gaussian-Moments Conjecture in all dimensions implies the Jacobian Conjecture; the two-dimensional case proved here is the first, and — by the counterexamples in *loc. cit.* — the last unconditionally available instance of that implication’s hypothesis.

Availability. The Lean sources build against `mathlib` and reproduce the axiom audit above. The proof was developed and machine-checked in the `TournamentH7` Lean project; the theorem `GMC2.gmc2` and its self-contained build target `GMC2Main` are the canonical entry points.

REFERENCES

- [1] H. Derksen, A. van den Essen, and W. Zhao, *The Gaussian moments conjecture and the Jacobian conjecture*, Israel J. Math. **219** (2017), no. 2, 917–928. [arXiv:1506.05192](#).
- [2] J. J. Duistermaat and W. van der Kallen, *Constant terms in powers of a Laurent polynomial*, Indag. Math. (N.S.) **9** (1998), no. 2, 221–231.

- [3] O. Mathieu, *Some conjectures about invariant theory and their applications*, in *Algèbre non commutative, groupes quantiques et invariants* (Reims, 1995), Sémin. Congr. **2**, Soc. Math. France, 1997, pp. 263–279.
- [4] W. Zhao, *Generalizations of the image conjecture and the Mathieu conjecture*, J. Pure Appl. Algebra **214** (2010), no. 7, 1200–1216.
- [5] W. Zhao, *Hessian nilpotent polynomials and the Jacobian conjecture*, Trans. Amer. Math. Soc. **359** (2007), no. 1, 249–274.
- [6] W. Zhao, *Images of commuting differential operators of order one with constant leading coefficients*, J. Algebra **324** (2010), no. 2, 231–247.
- [7] A. van den Essen, *Polynomial Automorphisms and the Jacobian Conjecture*, Progress in Mathematics **190**, Birkhäuser, Basel, 2000.
- [8] The mathlib Community, *The Lean mathematical library*, in *CPP 2020*, ACM, 2020, pp. 367–381.
Email address: `eliottcassidy2000@gmail.com`